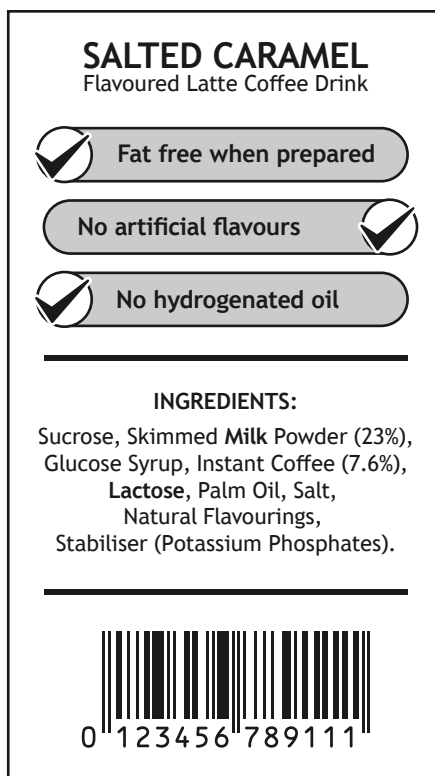
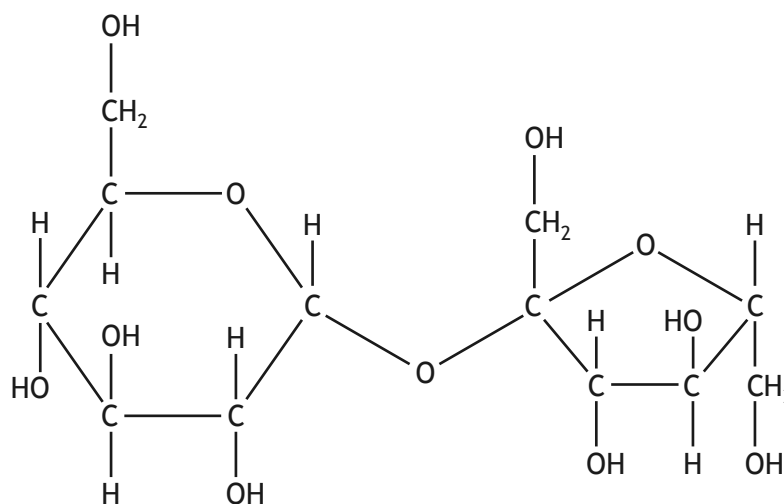


5. The label from a packet of instant latte coffee drink is shown below.

MARKS DO NOT
WRITE IN
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MARGIN



- (a) Sucrose is the main ingredient in the latte coffee drink. The structure of sucrose is shown.



Explain **fully** why sucrose is water soluble.

2



5. (continued)

- (b) Skimmed milk powder contains peptide molecules that are formed by joining together amino acid molecules.

Amino acids can be represented by the first three letters of their name; for example, Phe represents phenylalanine.

- (i) Hydrolysis of a peptide molecule found in milk gave a mixture of three smaller peptide molecules with the following amino acid sequences.

Tyr-Pro-Phe

Gly-Pro-Phe

Iso-Pro-Gly-Pro

The original peptide molecule was formed from seven amino acids joined together.

Write the amino acid sequence for the original peptide molecule.

1

_____ - _____ - _____ - _____ - _____ - _____ - _____

- (ii) Phenylalanine is an amino acid that cannot be made in the human body. State the term used to describe amino acids that the body cannot make.

1

- (c) Palm oil used in the latte coffee drink is an edible oil.

- (i) State the class of compounds to which edible oils belong.

1

- (ii) When palm oil is hydrolysed, mixtures of fatty acids and another compound are produced.

(A) Name the other compound produced.

1



* X 8 1 3 7 6 0 1 1 2 *

5. (c) (ii) (continued)

(B) The table gives information on the fatty acids obtained on hydrolysis of palm oil.

Fatty acid	Formula
Palmitic acid	$C_{15}H_{31}COOH$
Oleic acid	$C_{17}H_{33}COOH$
Linoleic acid	$C_{17}H_{31}COOH$
Linolenic acid	$C_{17}H_{29}COOH$

Name the saturated fatty acid obtained on hydrolysis of palm oil.

1

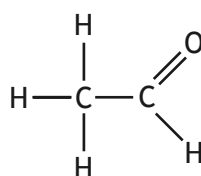
(iii) The latte coffee drink packet states it contains 'no hydrogenated oil'.
Hydrogenation takes place when unsaturated edible oils react with hydrogen.

State another name for this type of reaction.

1

(d) Coffee contains many different compounds.

(i) Acetaldehyde gives coffee a strong fruity aroma.



acetaldehyde

State the systematic name for acetaldehyde.

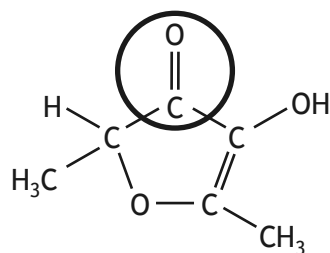
1



* X 8 1 3 7 6 0 1 1 3 *

5. (d) (continued)

(ii) Furanol gives coffee a caramel flavour.



furaneol

Name the functional group circled in furaneol.

1

(iii) Terpenoid molecules are also responsible for the aroma of coffee.

Terpenoids are classified by the number of isoprene units (2-methylbuta-1,3-diene) in the molecule.

Class of terpenoid	Number of isoprene units
Hemiterpenoid	1
Monoterpenoid	2
Sesquiterpenoid	3
Diterpenoid	4
Sesterterpenoid	5

(A) Cafestol, $C_{20}H_{28}O_3$, is a terpenoid found in coffee.

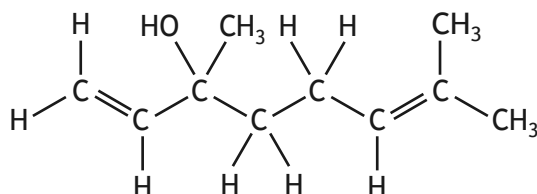
Name the class of terpenoids to which cafestol belongs.

1



5. (d) (iii) (continued)

(B) Linalool is a monoterpene found in coffee with the following structure.



Circle an isoprene unit on the linalool structure above.

1

(An additional structure, if required, can be found on *page 34*.)

- (e) One cup of the latte coffee drink contains 125.0 mg of caffeine. Consuming up to 5.7 mg of caffeine per kg of body mass per day is recommended to be safe for an adult.

Calculate the maximum number of cups of the latte coffee drink a 66 kg adult could safely drink without exceeding this recommendation.

1

[Turn over

